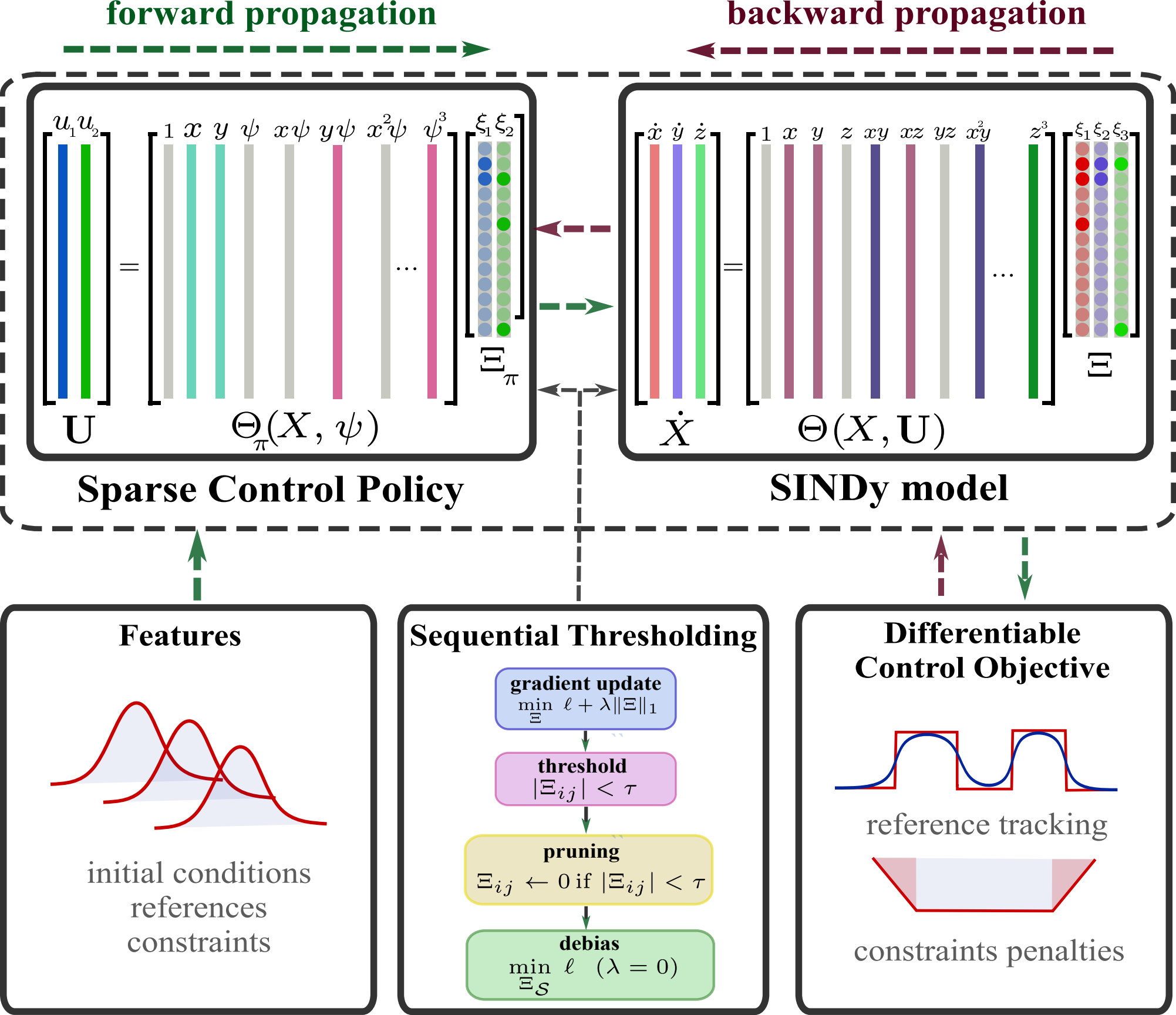

forward propagation
backward propagation
$u_1 u_2$
$1\ x\ y\ \psi\ x\psi\ y\psi\ x^2\psi\ \psi^3$
$\xi_1 \xi_2$
Ξ_π
U
$\Theta_\pi(X, \psi)$
$\dot{x}\ \dot{y}\ \dot{z}$
$1\ x\ y\ z\ xy\ xz\ yz\ x^2y\ z^3$
$\xi_1 \xi_2 \xi_3$
Ξ
$\dot{X}$
$\Theta(X, \mathrm{U})$
Sparse Control Policy
SINDy model
Features
initial conditions
references
constraints
Sequential Thresholding
gradient update
$\min_\Xi \ \ell + \lambda\|\Xi\|_1$
threshold
$|\Xi_{ij}| < \tau$
pruning
$\Xi_{ij} \leftarrow 0$ if $|\Xi_{ij}| < \tau$
debias
$\min_{\Xi_\mathcal{S}} \ \ell \quad (\lambda = 0)$
Differentiable
Control Objective
reference tracking
constraints penalties